Biodiversity, Conservation and its Environmental Issues

ASSIGNMENT 3 answers

Notes

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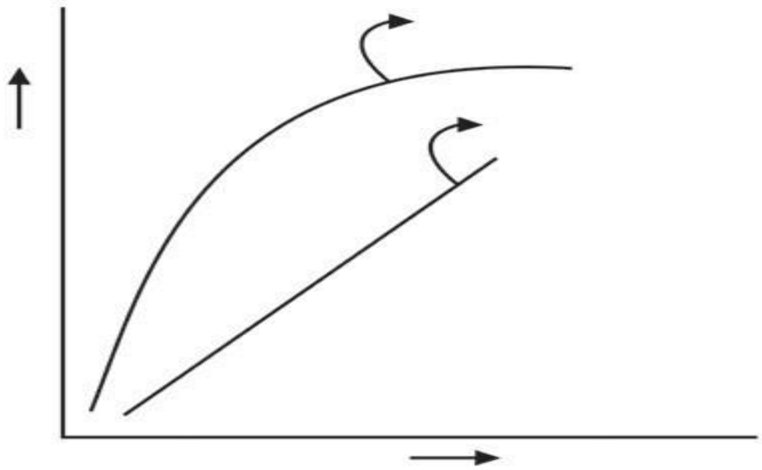
1. Observe the graph and select correct option. a. Line A represents,S=CA2



2. b. Line B represents, log C= log A + Z logS
3. c. Line A represents,S=CAZ
4. d. Line B represents, log S= log Z + C log A
5.

   6. Select odd one out on the basis of *Ex situ* conservation.
   a. Zoologicalpark
   b. Tissueculture
   c. Sacredgroves
   d. Cryopreservation
   Which of the following factors will favour species diversity?
   a. Invasive species b.Glaciation
   c. Forest canopy d. coextinction
   The term “terror of Bengal’ isusedfor______________.
   algal bloom b. water hyacinth c. increased BOD d. eutrophication

   CFC are air polluting agents which areproducedby______________.

   a. Dieseltrucks
   b. Jetplanes
   c. Rice field
   d. Industries

1. **Biodiversity**includes a vast array of species of microorganisms- viruses, algae, fungi, plants and animals occurring on Earth, either in terrestrial or aquatic habitat and the ecological complexes of which they arepart.
2. **Genetc**diversity is intraspecificdiversity.
3. **Species**diversity is interspecificdiversity.
4. **Species diversity** deals with variety of species (**species richness**) as well as numberof individuals of different species (**species evenness**) observed in area understudy.
5. Amphibian species diversity is more in **western**ghats than in easternghats.
6. Natural undisturbed tropical forests havemuch greater **species richness**than monoculture plantation of timber plant, developed by forestplantation.
7. **Ecological (Ecosystem) diversity**is related to the different types of ecosystems/ habitats within a given geographical area.
8. The diversity of life at all the three levels is now rapidly being modified by **modern man.**
9. There are two Biodiversity patterns viz**, a)Latitudinal and Altitudinal gradient and b)species-arearelationship.**
10. Biodiversity, barring Arid/ Semiarid and aquatic habitat, show **latitudinal and altitudinalgradient**.
11. Ithasbeenobservedthatspeciesrichnessis**high**atlowerlatitudesandthereisa**steady decline**towards the poles.
12. Factors like overall stability of tropical regions for millions of years, lesser climatic changes throughout the year and availability of plenty of sunlight that favoured**speciation.**
13. Lesser migrations in tropics might have reduced **gene flow**between geographically isolated regions and favouredspeciation.
14. Fruits being available throughout the year in rain forests, variety of**frugivorous** organisms is obviously more as compared to the temperate regions.
15. Species richness or diversity for plants and animals **decreases**as we move away from equator to thepoles.
16. Species diversity **is more**at lower altitudes than at the heigheraltitudes.
17. It is considered that number of species present is **directly**proportional to thearea.
18. German naturalist Alexander Von Humboldt observed that species richnessdoes **increase**with the increase in area but upto a limit.
19. For smaller areas, value of Z ranges between **0.1 to 0.2**regardless of species or region understudy.
20. Acommunityissaidtobe**stable**,ifaveragebiomassproductionremainsfairlyconstant over a period oftime.
21. Richdiversityleadstolesservariationinbiomassproductionoveraperiodoftime.This is called **Productivity- StabilityHypothesis**.
22. Paul Ehrlich, an ecologist from Stanford gave an analogy to explain significance of diversity. It is called **Rivet PopperHypothesis.**

23. There are three types of extinction viz, **natural**extinction, **mass**extinction and **man-made**(anthropogenic) extinction.
24. Great Indian Bustard alias *Maldhok*is a critically endangered bird. It is an exampleof **local**extinction due to habitat loss and hunting.
25. Dodo bird, stellar sea cow and passenger pigeon are few examples of extinction dueto **overexploitation.**
26. When a new species gets introduced into any ecosystem accidentally or intentionally, there are chances that it proves harmful for existing species. Sometimes, it can lead to extinction of local species. In such a case, it is called as **invasivespecies.**
27. **Predatorfish-Nileperch**inLakeVictoria,proveddeleteriousfor200**localspecies** of Cichlid fish.
28. InIndia,introductionofAfricancatfish***Clariasgariepinus***foraquaculturepurposehas proved harmful to endemic catfishvarieties
29. (IUCN)standsfor**TheInternationalUnionforConservationofNatureandNaturalResources .**
30. **Red Data Book**also known as Red List, is a book where conservation status of plant and animal species isrecorded.
31. **Extinct (EX)**, a designation applied to species in which the last individual has died or is not recorded
32. **ExtinctintheWild(EW),**acategorycontainingthosespecieswhosememberssurvive only incaptivity
33. **Critically Endangered (CR)**, a category containing those species that possess an extremely high risk of extinction with very few surviving members(50).
34. **Endangered (EN)**, a designation applied to species that possess a very high risk of extinctionasaresultofrapidpopulationdeclineof50tomorethan70percentoverthe previous 10 years (or threegenerations).
35. **Vulnerable (VU)**, a category containing those species that possess a very high risk of extinctionasaresultofrapidpopulationdeclineof30tomorethan50percentoverthe previous 10 years (or threegenerations).
36. **Near Threatened (NT)**, a designation applied to species that are close to becoming threatened or may meet the criteria for threatened status in the nearfuture.
37. **Least Concern (LC)**, a category containing species that are pervasive and abundant after carefulassessment
38. **Data Deficient (DD)**, a condition applied to species in which the amount of available data related to its risk of extinction, is lacking in someway.
39. **Not Evaluated (NE)**, a category used to include any of the nearly 1.9 million species described by scientists, but not assessed by theIUCN
40. **Bioprospecting**is systematic search for development of new sources of chemical compounds, genes, micro-organisms, macro-organisms, and other valuable products from nature.
41. Announcing Kanha forest as tiger reserve. This is called ***in situ*conservation.**
42. InMaharashtra,**Pawra**tribalsinSatpudahaveprotectedvarietiesofcornwithdifferent coloured kernels.
43. India has three of world's biodiversity hotspots (the areas with high density of biodiversity), **Western ghats**, **Indo-Burma** and**Eastern-Himalayas**.

44. In many cultures, stretches of forests were set aside and protected in the name of Almighty, which are called **sacredgroves**.

45. Animals which have decreased in number, are allowed to breed in captivity in order to protect them.Eg**crocodile bank ofChennai**.

46. Moderntechniquesliketissueculture,*invitro*fertilizationofeggsandcryopreservation (preservation at low temperature -1960C) of gametes, are used to protect **endangered**species.

47. Air pollutants are of two types – **particulate**pollutants and **gaseous**pollutants.

48.**Carbonmonoxide(CO)**isapoisonousgasproducedbyincompletecombustionof fuel such as coal or wood.

49. NO2 when combines with water vapours forms **nitricacid.**

50. **Electrostaticprecipitator**ismostwidelyusedforremovingparticulatematterlikesoot and dust present in industrialexhaust

51. **Electrostatic precipitator**can remove almost 99% particulate matter present in exhaust from a thermal power plant.

52. **Exhaust gas Scrubbers**are used to clean air for both dust and gases by passing it through dry or wet packingmaterial.

53. CNG stands for **(compressed naturalgas).**

54.(BS) stands for **Bharat stage emission standards**

55. Exposure to extremely high sound level (**150 decibels or more**) generated during a take-off of a jet plane or rocket, may damage ear drums and cause permanent hearing loss.

56. Itispossibletoestimatebiodegradableorganicmatterinsewagewaterbymeasuring **Biochemical Oxygen Demand (BOD)**.

57. **Biochemical Oxygen Demand (BOD)**is the amount of dissolved oxygen required by microorganisms for decomposing the organic matter present inwater.

58. **Biochemical Oxygen Demand (BOD**is expressed in milligram of oxygen per litre (mg/L) ofwater.

59. **High BOD i**ndicates intense level of microbialpollution.

60. Presenceoflargeamountofnutrientsinwatercausesexcessivegrowthof**planktonic** free-floating algae specially, blue green algae

61. Another threat to aquatic ecosystem, is water hyacinth (***Eichhorniacrassipe***s).

62.**Natural Eutrophication**is aging of a lake due to nutrient enrichment ofwater.

63. However,duetopollutantsfromhumanactivitieslikeeffluentsfromagriculturallands, industries and homes (house hold) have accelerated this aging process. This phenomenon is called **Cultural or AcceleratedEutrophication.**

64. **Ecological sanitation (Ecosan)**is an approach to sanitation provision which safely reuses excreta in agriculture. It reduces the need for chemicalfertilizers.

65. **Sanitary landfills**are substitute for open burning dumps.

66. Irreparable computers and other electronic goods as well as electrical waste are known as **ewastes.**

67. Withoutgreenhouseeffect,theaveragetemperatureofEarthwouldhavebeen**-18^{0}C** _____ rather than average of 15^{0}C.

68. Hence CO2 and CH4 are commonly called **greenhouse gases**.

69.Thinned ozone layer, commonly called **ozonehole.**

70. Thickness of the ozone in a column of air from the ground to the top of theatmosphere is measured as **<u>Dobson units(DU).</u>**
71. Recognising the harmful effects of ozone depletion, an international treaty, known as the**<u>MontrealProtocol</u>**wassignedatMontreal(Canada)in1987tocontrolemissionof ozone depleting substances
72. **<u>Slashandburnagriculture</u>**commonlycalled**JhumCultivation**innortheasternparts of India, where farmers cut down trees of the forest and burn the plantremains..
73. A Bishnoi woman **<u>Amrita Devi</u>**hugged the trees and lost her life in an attempt for protecting theforest.
74. In 1980s, realising the significance of participation of local communities, Government of India introduced the concept of **<u>Joint Forest Management</u>**(JFM) so as to work closely with local communities for protecting and managingforests.

<u>Q. 2 Very short answer</u>

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1. Give two examples of biodegradable materials released from sugarindustry.

Ans: A sugar mill produces several wastes us as molasses, **bagasse**, filter press cake, waste water, **bagasse ash**

2. Name any 2 modern techniques of protection of endangeredspecies.

Ans: Now a days, modern techniques like tissue culture, *in vitro* fertilization of eggs and cryopreservation (preservation at low temperature -1960C) of gametes, are used to protect endangered species.

3. Where was ozone hole discovered?

Ans: Antarctic ozonehole

4. Give one example of naturalpollutant.

Ans: Anaturalpollutantisapollutantcreatedbysubstancesofnaturaloriginsuchasvolcanic dust, sea salt particles, photochemically formed ozone, and products of forest fibres, among others.

5. What do you understand by EW category of livingbeing?

Ans: <u>Extinct in the Wild (EW),</u> a category containing those species whose members survive only in captivity, they are extinct in their natural free habitat.

<u>Q. 3 Short answer type questions.</u>

1. *Dandiyaraas*is not allowed after 10.00 pm.Why?

Ans: 1. In India, the Air (Prevention and control of pollution) Act 1981, amendment 1987, includes **noise** as an air pollutant.

2. Noiseisanundesiredhighlevelofsound.Noisecausespsychologicalandphysiological changes in human beings.

3. There is dire need of creating awareness about noise pollution caused during festivals and processions in our society.

4. Noise also can cause sleeplessness, increased heartbeat, altered breathing pattern and psychologicalstress.

5. Noise may negatively interfere with child’s learning and behaviourpattern.

6. Supreme court of India has banned loudspeakers at public gatherings after10pm.

7. **Tropical regions exhibit more species richness as compared to polar regions. Justify.**

Ans: 1. Tropical regions for millions of years, lesser climatic changes throughout the year and availability of plenty of sunlight that favouredspeciation.

2. Tropical areas have less often experienced drastic disturbances like periodic glaciations observed at poles

3. Lesser migrations in tropics might have reduced gene flow between geographically isolated regions and favouredspeciation.

4. Scientists also have considered availability of more intense sunlight, warmer temperatures and higher annual rainfall in tropics, as factors responsible for bountifulness of theseregions

- 5. In more or less constant climatic conditions and abundance of resources, some animals enjoy foodpreferences.
- Fore.g.fruitsbeingavailablethroughouttheyearinrainforests,varietyoffrugivorous organisms is obviously more as compared to the temperateregions.

8. **How does genetic diversity affect sustenance of aspecies?**

➢ **Ans:**

1. It is the intraspecificdiversity.

2. Itisthediversityinthenumber andtypesofgenesaswellaschromosomespresent in different species and also the variation in the genes and their alleles in the same species.

3. Existence of subspecies, races are examples of geneticdiversity.

4. Greater the diversity, better would be sustenance of aspecies.

5. You know about 1000 varieties of mangoes and 50,000 varieties of rice or wheat in India.

6. Genetic variations (e.g. allelic genes) lead to individual differences withinspecies.

7. Such variations pave way toevolution.

8. They also improve chances of continuation of species in the changing environmental conditions or allow the best adapted to survive

9. Example: Medicinal plant *Rauwolfiavomitoria which secretes active component reserpine, is found in different Himalayan ranges. This plant shows variations in terms of potency and concentration of active chemical, from location tolocation.*

10. Green house effect is boon or bane? Give youropinion.

Ans: **<u>Green House effect</u>**

1. It is responsible for heating of earth's surface andatmosphere.
2. Without greenhouse effect, the average temperature of Earth would have been -180C rather than average of150C.
3. Of the solar radiation that reaches earth, Clouds and gases reflect about 1/4th and absorb some ofit.
4. But half of total incoming radiations reach the earth's surface and heatit.
5. Small portion of it, is reflected back. Earth's surface re emits the heat in the form of infraredradiations.

6. Part of these radiations do not escape into the space because atmospheric gases (CO2, CH4) absorb a majorportion
7. The molecules of these gases radiate heat energy and a major part of it comes back to earth'ssurface.
8. Thus reheating theearth.
9. Thiscycleisrepeatedmanytimes.HenceCO2andCH4arecommonlycalled **greenhouse gases**.

Boon of green house effect

1. the earth is warm enough. Simply because the greenhouse gases are able to entrap the solar radiation and bounce it back to the earth's surface, is the reason the earth hasn't frozen yet.
2. The greenhouse gases also ensure that we are not fried by the sun's heat by absorbing some percentage of the radiation.
3. Therefore,ontheonehand,benefitsof greenhousegasesturntotheirabilitytoprevent theheatfromescapingtheearth'ssurfacehelphumanlifehaveawarmenoughclimate.
4. Ontheotherhand,theirabilitytobounceofafairshareoftheradiation,helpthehuman life avoid an unpleasantly hot and unbearableatmosphere.
5. importance in maintaining our planet's waterlevel.
6. In the absence of greenhouse gases, the polar caps would rapidly melt, raising water levels beyond alarmingmarks.

Bane of green House effect

1. GlobalWarming
2. Increasing WaterLevels
3. Destruction of Marine Life

My opinion

Advantages and disadvantages of greenhouse effect can sensitize students about the need of the hour and help them realize how small steps they take can go a long way to preventing the increase in the greenhouse effect. In my opinion excess green house effect should be prevented.

11. **How does CO cause giddiness andexhaustion?**

    Ans: 1. Carbon monoxide poisoning occurs when carbon monoxide builds up in your bloodstream.

    2. When too much carbon monoxide is in the air, your body replaces the oxygen in your red blood cells with carbon monoxide.

    3. This leads to giddiness andexhaustion.

    4. This eventually can lead to serious tissue damage, or evendeath.

12. **Name two types of particulate pollutants found in air. Add a note on ill effects of the same on human health.**

Ans: Types of Particulate Air Pollutants:

Smoke,smog,pesticides,heavymetals,dustandradioactiveelementsaretheexamples of particulate pollutants

<u>Ill Effects</u>:

1. AccordingtoCentralPollutionControlBoard(CPCB),particulatematterofsize2.5µm or less in diameter (PM2.5) are responsible for causing the greatest harm tohumans.
2. These fine particulates can be inhaled deep into the lungs and are responsible for irritation, inflammation and damage tolungs.
3. In addition to this, it causes breathing and respiratory disorders and prematuredeaths.

<u>Q. 4 Long answer type questions.</u>

1. Montreal protocol is an essential step. Why is itso?

Ans: 1. **Ozone** is a form of oxygen.In the stratosphere, ozone is photo-dissociated and is generated by absorption of short wave length UV radiations.

$$O_3 \longrightarrow O_2 + [O]$$

$$O_2 + [O] \xrightarrow{UV\ RAYS} O_3$$

2. This ozone layer acts as shield that absorbs UV radiations from the sun and protects all types of life onearth

3. Thickness is more above the poles than at theequator.

4. UV rays are highly injurious to living organisms since DNA and proteins of living organisms absorb UV rays and its high energy breaks the chemical bonds within these molecules.

5. Thickness of the ozone in a column of air from the ground to the top of the atmosphere is measured as **Dobson units**(DU).

6. The ozone shield has been disturbed due to increased rate of ozone degradationby **Chlorofluorocarbon** (CFC).

7. UV rays act on them and release Cl atoms. Cl degrades ozone releasing molecularoxygen. Cl atoms act as catalyst.So

8. Thishasresultedinformationoflargeareaofthinnedozonelayer,commonlycalled**ozone hole.**

9. It causes aging of skin, damage to skin cells and various types of skin cancers. In human eye,corneaabsorbsUV-Bradiationsandahighdoseof UV-Bcausesinflammationofcornea called **snow blindness,** cataract etc; Such exposure may permanently damagecornea.

10. Recognising the harmful effects of ozone depletion, an international treaty, known as the **Montreal Protocol** was signed at Montreal (Canada) in 1987 to control emission of ozone depleting substances.

11. Latermanymoreeffortshavebeenmadeandprotocolshavelaiddowndefiniteroadmaps separately for developing and developed countries for reducing emission of CFCs and other ozone depleting chemicals

2. **Name any 2 personalities who have contributed to control deforestation in our country. Elaborate on importance of their work.**

Ans:I) <u>SaalumaraThimmakka</u>

1. Best example of peoples participation in reforestation is **SaalumaraThimmakka,** an Indian environmentalist from state of Karnataka noted for her work in planting and tendingto385banyantreesalonga4kmstretchofhighwaybetweenHulikalandKudur.

2. She has also planted nearly 8000 other trees. Her work has been honoured with the National Citizens Award of India. She was also conferred with Padma Shri in2019.

II**)<u>MoirangthemLoiya</u>**

1. MoirangthemLoiyafromManipurdedicated17yearsofhislifetorestorePunshilokforest.
2. He left his job and took over the task of bringing back the lost glory of 300 acres forest land.
3. He planted a variety of trees like, bamboo, oak, ficus, teak, jackfruit andmagnolia.
4. Today the forest has over 250 varieties of plants including 25 varieties of bamboo. It is now selected as home by great diversity of animalstoo.

3. How BS emission standards changed over time? Why is it essential?
Ans:

Vehicle	**Norms**	**Cities of Implementation**
4 wheelers	Bharat Stage II	All metro cities
4 wheelers	Bharat Stage III	Throughout the country since October 2010
4 wheelers	Bharat Stage IV	13 mega cities (Delhi and NCR, Mumbai, Kolkata, Chennai, Bengaluru, Surat, Kanpur, Agra, Lucknow, Solapur) since April 2010.
2 wheelers	Bharat Stage III	Throughout the country since October2010
3 wheelers	Bharat Stage III	Throughout the country since October2010

1. Have you noticed that BS V is missing in abovetable?
2. Note that, in 2001, Bharat stage II emission norms were set for CNG and LPG vehicles.
3. It stipulates that emission of sulphur be controlled at 50 ppm in diesel and 150ppm in petrol.
4. Aromatic hydrocarbons should be just 42% in concernedfuel.
5. The aim is to reduce sulphur emission to 50ppm in petrol and diesel along with aromatic hydrocarbons to35%.
6. Government of India directly adapted BS VI in the year 2018, skipping BSV.
7. These efforts decreased the levels of CO2 and SO2 inDelhi.

4. How Indian culture and traditions helped in bio-diversity conservation? Give importance of conservation in terms of utilitarianreasons.
Ans: <u>Sacred Grooves</u>:
Indiancultureandtraditionsarealwaysconnectedwithnatureandritualsarelaiddown to protect biodiversity. In many cultures, stretches of forests were set aside and protectedinthenameofAlmighty,whicharecalled**sacredgroves**.Suchsacredgroves

arefoundinKhasiandJaintiahillsinMeghalaya,WesternghatregionsofMaharashtra and Karnataka, Aravali hills of Rajasthan and Bastar, and Chanda and Sarguja areas of MadhyaPradesh.

- . **Broadly utilitarianreasons**
  1. If we find out the cost of oxygen cylinder and try to calculate the value of oxygen we breathe with such ease; we will understand what nature is giving us forfree!
  2. Animals play a crucial role in pollination and seeddispersal.
  3. Amazon forest is estimated to produce 20% of total oxygen of earths atmosphere. We need to consider recreational use of diversitytoo.
  4. You must have come across the news about devastating fires in amazon rainforest in August 2019.
  5. These are mainly caused in Brazil and are more manmade thannatural.
  6. The slashand burn policy of locals to reclaim forestland has caused a towering 906000 hectares of forest devastation, only in the year2019.

Application based Questions

1. Why should we have national parks and sanctuaries?

Ans: What are the Benefits of National Parks and Wildlife Sanctuaries?

1. **National Parks Help ProtectWildlife**

Unfortunately many animal species today face extinction, mainly because their natural habitats are being steadily destroyed. National parks safeguard these habitats, and provide a safe space for wildlife to breed and survive..

2. **National Parks Help ProtectLandscapes**

Animals are not the only things that are at risk of disappearing. Landforms like mountains, rainforests, gorges and dunes are at risk of disappearing if they are not protected from the actions of humans and also the natural action of the environment. Many landforms are at risk from pollution, and when they are controlled under national park status they have a better chanceofsurvival.Landformsinnationalparksareprotectedfromdevelopment,destruction, and pollution.

3. **Parks and Sanctuaries PreserveHistory**

Historical structures built on national park land are preserved in order to give us a better idea of how people lived in the past, and how their cultures worked. There are many different structures that can be preserved which allow people to learn from the past and continue building for the future.

4. **Helping Preserve Cultures andTribes**

In many national parks around the world people live generally apart from main civilization, and their culture and members are largely protected thanks to the status of the national park. National parks not only protect animals and wildlife, they can also sometimes protect people too.

5. **Giving People the Chance for Healthy Activity**

National parks and to some extent wildlife sanctuaries also exist to provide members of the public with the space for healthy exercise and recreation in the open air. It is important to conserve places where the natural environment is intact, so that people can slow down, enjoy nature, and get some exercise by walking, running, or riding bikes

2. **What is *in situ* and *ex situ*conservation?**

- **Ans:** In situConservation

  1. Protection of natural habitat is protected. e.g. Announcing Kanha forest as tiger reserve. This is called *in situconservation*
  2. Around 34 Biodiversity hotspots have been identified by the conservationists. (regions with high species richness & density.) These areas need to be protected strategically by setting legislative measures apart from awareness and conservation.
  3. *Insituconservationalsoincludesintroductionofvarietiestraditionallyusedinto farming and horticulture. E.g. In Maharashtra, Pawratribals in Satpuda have protected varieties of corn with different colouredkernels.*
  4. India has three of world's biodiversity hotspots (the areas with high density of biodiversity), Western ghats, Indo-Burma and Eastern-Himalayas. It has been estimated that protection of these diversity rich hotspots could reduce extinction rate by almost 30%.

  5.India, at present has 14 biosphere reserves, 90 national parks and 448 wildlife sanctuaries. In Maharashtra, there are 5 national parks and 11sanctuaries.

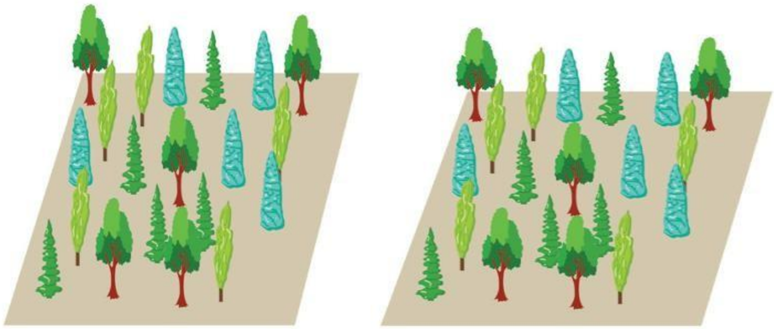
6. In many cultures, stretches of forests were set aside and protected in the name of Almighty, which are called sacred groves. Such sacred groves are found in Khasi and JaintiahillsinMeghalaya,WesternghatregionsofMaharashtraandKarnataka,Aravali hills of Rajasthan and Bastar, and Chanda and Sarguja areas of MadhyaPradesh.

<u>Ex-SITU conservation</u>

- Sometimes when a species is critically endangered, special measures have to be under taken to protect it. It might be protected in captivity, as one of the measures of protection. This is called *Ex situconservation*
- Wild life safari parks, zoological parks and botanical gardens serve this purpose. Animals which have decreased in number, are allowed to breed in captivity in order to protect them.Eg crocodile bank ofChennai.
- Seed banks are established to conserve wild varieties of food grains andvegetables.

  Now a days, modern techniques like tissue culture, *in vitro fertilization of eggs and cryopreservation (preservation at low temperature -1960C) of gametes, are used to protect endangered species.*

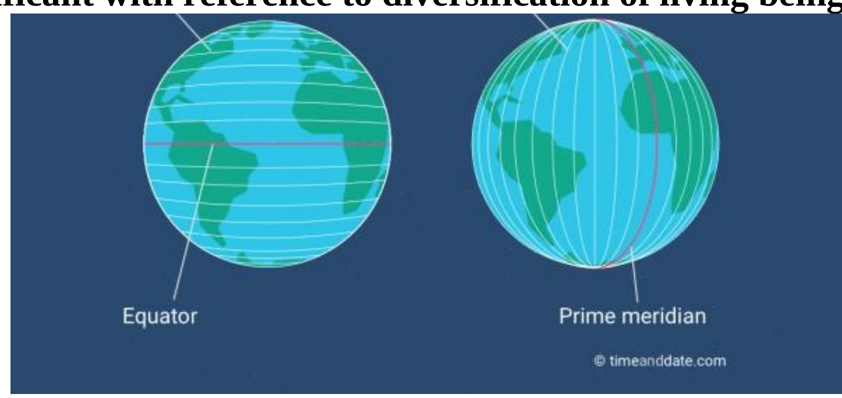
3. **What can you say about species diversity A andB?**



A **B**

Ans: 1. Both in section A and section B have species richness and species evenness
2. There are 4 different types of trres in both section hence they are species rich.
4. have equal number of species hence there is speciesevenness.

4 What are latitudes and longitudes? Which of these imaginary lines are more significant with reference to diversification of living beings?Why?

Ans: 

1. **Latitudes:** Latitudes are imaginary parallel horizontal lines on the earth's surface.
2. **Longitudes:** Longitudes are imaginaryparallelvertical lines on the earth's surface
3. **Latitudes are more significant with reference to diversification of living beings.**
4. Ithasbeenobservedthat speciesrichnessishighatlowerlatitudesandthereisa steady decline towards thepoles

5. In short, species richness or diversity for plants and animals decreases as we move away from equator to thepoles.
6. It is maximum in tropical rain forests e.g. Amazon rain forest (40,000 plants, 1300 birds, 427 mammals, 3000species).
7.

5. What is biodiversity? Explain genetic diversity with suitableexample.

Ans: Biodiversity:

- Diversity is variety. This variety of life is called biodiversity. Biodiversity includes a vastarrayofspeciesThediversityiswithrespecttosize(microscopicto macroscopic), shape, colour, form, mode of nutrition, type ofhabitat, reproduction, motility, duration of life cycle span, etc
- **Geneticdiversity**
- It is the intraspecificdiversity.
- It is the diversity in the number and types of genes as well as chromosomes present in differentspeciesandalsothevariationinthegenesandtheirallelesinthesamespecies
- **Example**
- Existence of subspecies, races are examples of geneticdiversity.
- Greater the diversity, better would be sustenance of aspecies.
- You know about 1000 varieties of mangoes and 50,000 varieties of rice or wheat in India.

6. Species richness goes on decreasing as we move from equator to pole. Explain

Ans: 1. It has been observed that species richness is high at lower latitudes and there is a steady decline towards thepoles.

2.Tropical regions for millions of years, lesser climatic changes throughout the year and availability of plenty of sunlight that favouredspeciation

7. Tropicalareashavelessoftenexperienceddrasticdisturbanceslikeperiodicglaciations observed at poles
8. Lesser migrations in tropics might have reduced gene flow between geographically isolated regions and favouredspeciation.
9. Scientists also have considered availability of more intense sunlight, warmer temperatures and higher annual rainfall in tropics, as factors responsible for bountifulness of theseregions
10. In more or less constant climatic conditions and abundance of resources, someanimals enjoy foodpreferences.
11. Fore.g.fruitsbeingavailablethroughouttheyearinrainforests,varietyoffrugivorous organisms is obviously more as compared to the temperateregions.
12. .In short, species richness or diversity for plants and animals decreases as we move away from equator to the poles.
13. Itismaximumintropicalrainforestse.g.Amazonrainforest(40,000plants,1300birds, 427 mammals, 3000species

8. What is ballast water and how can it bring about introduction of alien species into any ecosystem?

Ans: Ballast water :

Ballast water is essential for the safe operation of **ships**. It is used to adjust theoverall weightofthe**vessel**anditsinternaldistributioninordertokeepthe**ship**floatingsafely, upright and in a stablecondition.

Introduction of Alien species:

Thedispersalofinvasivespeciesbyballastwaterreferstotheunintentionalintroduction ofinvasivespeciestonewhabitatsviatheballastwatercarriedbycommercialshipping vessels. Ballast water spreads an estimated 7000 living species to new habitats across the globe.

7. Explain how loss of species diversity can harm ecosystem?

Ans:

1. Loss of biodiversity leads to the overall imbalance in theecosystem.
2. The chief serious aspect of loss of biodiversity is extinction of species. There are three types of extinction viz, **natural** extinction, **mass** extinction and **man-made** (anthropogenic)extinction.

3.Damage to biodiversity takes place due to both, natural and manmade reasons. Natural reasons include forest fires, earthquakes, volcanic eruptions etc. Manmade reasons are habitat destruction, hunting, settlement, overexploitation andreclamation.

4. The current loss of biodiversity is considered to be the Sixth extinction which is progressing at an alarming rate which is estimated to be 100 to 1000 times faster than prehumantimes.

5. Ecologists blame this to the human intervention in naturalhabitats.

6. Theydonotforgettowarnthatifthecurrenttrendcontinues,wemightloseabout50% ofdiversity.

7. Loss of biodiversity in any area can lead to the decline in plant production, lower resilience to environmental disturbance likeflood.

8. It may also lead to alteration in environmental processes like disease cycles, plant productivityetc.

8. Give various categories of endangered species explained by IUCN

The International Union for Conservation of Nature and Natural Resources (IUCN) maintains a Red Data Book also known as Red List, where conservation status of plant and animal species is recorded**.** They are as follows:-

1. **Extinct (EX)**, a designation applied to species in which the last individual has died or is not recorded.
2. **Extinct in the Wild (EW)**, a category containing those species whose members survive only incaptivity
3. **Critically Endangered (CR)**, a category containing those species that possess an extremely high risk of extinction with very few surviving members(50).
4. **Endangered (EN)**, a designation applied to species that possess a very high risk of extinction as a result of rapid population decline of 50 to more than 70 percent over the previous 10 years (or threegenerations).
5. **Vulnerable (VU)**, a category containing those species that possess a very high risk of extinction as a result of rapid population decline of 30 to more than 50 percent over the previous 10 years (or three generations).
6. **Near Threatened (NT)**, a designation applied to species that are close to becoming threatened or may meet the criteria for threatened status in the nearfuture.

7. **Least Concern (LC)**, a category containing species that are pervasive and abundant after careful assessment
8. **DataDeficient(DD)**,aconditionappliedtospeciesinwhichtheamountofavailabledata related to its risk of extinction, is lacking in someway.
9. **Not Evaluated (NE)**, a category used to include any of the nearly 1.9 million species described by scientists, but not assessed by theIUCN

9. What do you understand by invasive species? How does it affect local population?

Ans: Alien species invasion

When a new species gets introduced into any ecosystem accidentally or intentionally, there are chances that it proves harmful for existing species.

- Sometimes, it can lead to extinction of localspecies.
- In such a case, it is called as invasive species. E.g. the carrot grass (*Parthenium), Lantana and water hyacinth (Eichhornia). Introduction of predator fish - Nile perchin Lake Victoria, proved deleterious for 200 local species of Cichlidfish.*
- InIndia,introductionofAfricancatfish*Clariasgariepinusforaquaculturepurposehas proved harmful to endemic catfishvarieties.*

One of the major reasons of such a harmful effect of alien species is, lack of local predator

10. What are sacredgroves?

Ans: **<u>Sacred Grooves</u>**:

1. Indianculturendtraditionsarealwaysconnectedwithnatureandritualsarelaid down to protect biodiversity.
2. In many cultures, stretches of forests were set aside and protected in the nameof Almighty, which are called **sacredgroves**.
3. Such sacred groves are found in Khasi and Jaintia hills in Meghalaya, Western ghatregionsofMaharashtraandKarnataka,AravalihillsofRajasthanandBastar, and Chanda and Sarguja areas of MadhyaPradesh

11. What is pollution? Enlist its types.

Ans: Pollution:Pollutionistheintroductionofcontaminantsintothenaturalenvironmentthat cause adverse change. Pollution can take the form of chemical substances or energy, such as noise, heat or light. Pollutants, the components of pollution, can be either foreign substances/energies or naturally occurringcontaminants

The different types of pollution include:

- Air pollution.
- **Waterpollution**.
- **Soilpollution**.
- Radioactivepollution.
- Noisepollution

12. Define pollutant. How are our daily activities responsible forpollution?

Ans: A pollutant is a substance or energy introduced into the environment that has undesired effects, or adversely affects the usefulness of a resource.

Daily activities responsible for pollution

- Industrial and vehicularemissions

1. Heavy industries and vehicles act as major sources of primary air **pollutants**. Industries like energy, shipping, manufacturing, and automobiles discharge air **pollutants** such as fine particulate matter, volatile organic compounds, nitrogen and carbon compounds, ground-level ozoneetc
2. Humans around the world litter **every day**. As a result, most **water** becomes **polluted** by human **activities**. These include: throwing sewage and industrial waste into rivers and oceans, using harmful fertilizers, and even littering. All of these human **activities** affect our **water**sources
3. Soil polltion**: Activities** such as stock breeding and intensive farming use chemicals, pesticides and fertilisers that pollute the **land**, just as happens with heavy metals and other natural and man-made chemicalsubstances.
4. Noise pollution: Some of the main sources of **noise** in residential areas include loud music, transportation (traffic, rail, airplanes, etc.), lawn care maintenance, construction, electrical generators, explosions, andpeople.

13. Differentiate between *in situ and ex situ*conservation.

Ans:

***In situ* conservation**	***Ex situ* conservation**
This method involves protection of endangered species in their natural habitat.	It involves placing of threatened animals and plants in special care unit for their protection.
It helps in recovering populations in the surroundings where they have developed their distinct features.	It helps in recovering populations or preventing their extinction under stimulated conditions that closely resemble their natural habitats.
e.g. national parks, biosphere reserves, wildlife sanctuaries, etc.	e.g. botanical garden, zoological parks.

14. Write a note on BD Act 2002.

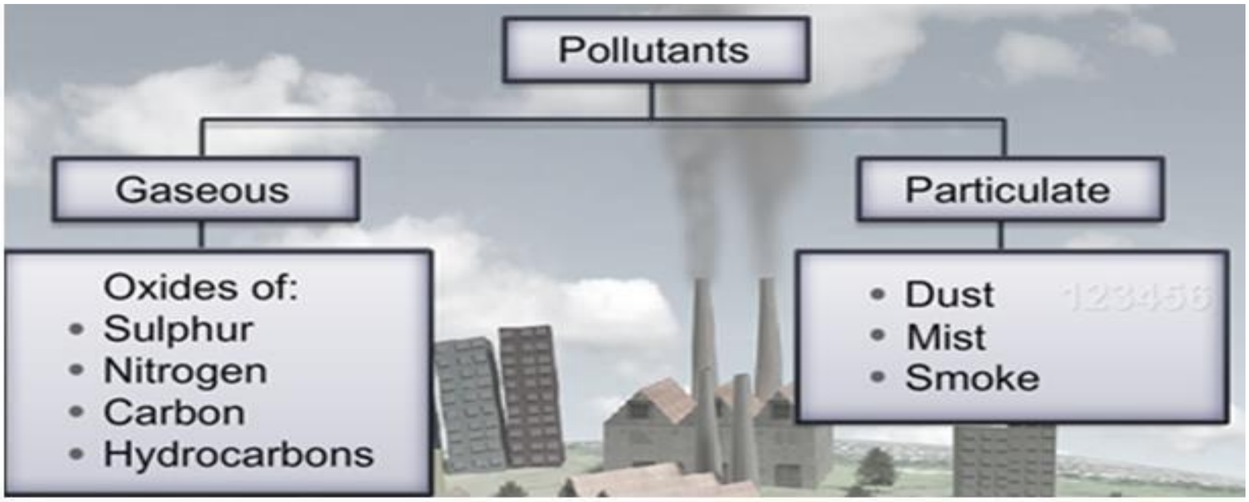
Ans:

1. India participated in Earth Summit, Rio de Janeiro and is a party to Convention on Biological Diversity(CBD-1992).
2. In order to provide framework for the sustainable management andconservation of our country's natural resources, government passed Biological Diversity Act (BD Act) in the year 2002 in compliance withCBD.
3. Thelawbroadlydefinesbiodiversity,asplants,animalsandmicroorganismsand their parts, their genetic materials and by-products. It excludes value added products and human geneticmaterial.
4. Regulation of access to Indian biological resources as well as scientific cataloguing of traditional knowledge about ethnobiological materials, were the main objectives for proposing thisact.
5. Athree-tiersystemhasbeenestablishedwith**NationalBiodiversityAuthority**(NBA) at the national level, the **State Biodiversity Boards** (SBBs) at the state level, and **Biodiversity Management Committees** (BMCs) at the local level for approval of utilization of any biological resource for commercial or researchpurpose.

6. It is mandatory for foreigners, NRIs as well as Indian citizens and institutions to seek permission from NBA before exploiting localresource.
7. NBA has powers of civil court. Not seeking approval of NBA, can incur jail and fine up to 10 lakhrupees.

15. Does particulate matter help to reduce atmospherictemperature?
Ans: yes

16. Describe any 2 particulate and gaseouspollutants.
Ans:



17. Explainvarioustechnologiesusedforremovingparticulatematterfromdifferent sources of air pollution.

✓ **Ans: <u>Electrostaticprecipitator</u>**

- It is most widely used for removing particulate matter like soot and dust present in industrialexhaust.Itcanremovealmost99%particulatematterpresentinexhaustfrom a thermal powerplant.
- Inthismethod,highvoltageisappliedandelectricdischargetakesplacewhichcauses ionisation of air in thesmokestack.
- Free electrons in the ionised air get attached to the gaseous/dust particles moving up thestack.
- Thesenegativelychargedparticlesmovetowardsthepositiveelectrodeandsettledown there.
- These particles are removed by vibrations of the electrodes and collected in the reservoir.

<u>Exhaust gas Scrubbers</u>

- They are used to clean air for both dust and gases by passing it through dry or wet packing material.
- It can remove gases like SO2. In the scrubber, the exhaust is passed through a spray of water orlime

<u>Catalytic converters</u>:

- Motor vehicles equipped with catalytic converter should use unleaded petrol because lead in the petrol, inactivates thecatalyst

18. Whataretheilleffectsofnoisepollutiononhumanhealth?4.Giveanynormsfor reducing sulphur and aromatic contents of petrol anddiesel.

Ans: Ill effects of noise pollution on human health

- :-Noise also can cause sleeplessness, increased heartbeat, altered breathing pattern and psychological stress. Noise may negatively interfere with child's learning and behaviourpattern.Thecommonsourcesofnoisepollutionaremachines,transportation, construction sites, industries etc.
- Govt.ofIndiahasrulesandregulationsagainstfirecrackersandloudspeakers.Supreme court of India has banned loudspeakers at public gatherings after10pm.
- **Norms for reducing sulphur and aromatic contents of petrol anddiesel.**
    1. According to new fuel policy, the norms are set to reduce sulphur and aromatic content of petrol and diesel. Another provision is upgradationofengines.
    2. For this, **Bharat stage emission standards (BS)** are set. These standards are equivalent to Euro norms and have evolved on similar lines as Bharat Stage II (BS II) to BS VI from 2001 to2017.

19. **Find out different sources of noise pollution in yoursurrounding.**
Ans: Noise pollution: Some of the main sources of **noise** in residential areas include loud music, transportation (traffic, rail, airplanes, etc.), lawn care maintenance, construction, electrical generators, explosions, and people

20. **Where does domestic waste water go in urban and ruralareas?**
Ans: Domestic waste water from urban and rural areas goes in sewage treatment plant.
21. **What is importance of sewage treatmentplant?**
Ans: i. The procedure for removing contaminants from the **wastewater** basically from the household **sewage** is called **sewage treatment**.
ii. It has to undergo the chemical, physical and biological procedure to remove these contaminants and give out an environmentally safe **treatedeffluent**.
iii. Evenasmallquantityofabout0.1%impuritiesinwater,canmakeitunfitforhuman consumption. Solids are relatively easy to separate but dissolved salts such as nitrates, phosphates, other nutrients and toxic metal ions as well as organic compounds, are difficult to remove.
Hence sewage treatment plants are necessary in our civilization.
22. **We have studied about health effects of noise pollution on humans. How does this pollution affect birds? Does it affect marine life? Findout.**
Ans: <u>Effects of noise pollution on marine animals</u>

i. **M**any birds are affected by thisnoise
ii. Some species are very sensitive to noise, they are easily frightened and are under constant stress that affects their general performance andbehavior,"
iii. Noise frightens birds, creating stressful situations in which they fly away using energytheycouldhaveusedtosearchforfood,torestorforothercrucialactivitieslike matingv
iv. "Male birds sing to attract mates and their song determines whether they arechosen by a female bird or not,"
v. Noise pollution, however, affects bird communication since they have to compete with city noise, having to emit louder and higher frequency sounds, which make them consume moreenergy.

vi. Territorial birds are also affected since they use sounds to defend theirterritories.

Effects of noise pollution on marine animals

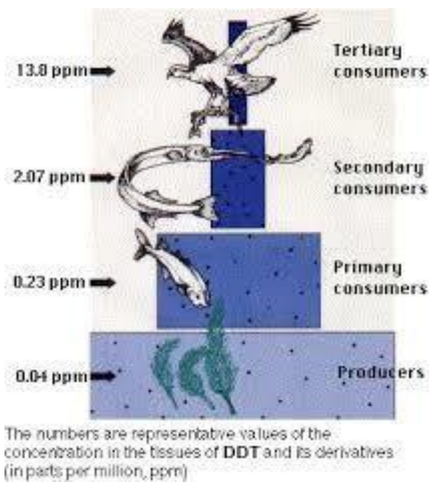
i.Noise means stress and impairs the **animals**' immune system which makes them more susceptibletoillnessingeneral.Ii.**Oceannoisepollution**alsocauses**marineanimals**toflee andabandonvaluablehabitats,eitherbecauseofdirect**impact**orbecausetheyhavetofollow their fleeingprey.

23. **Define water pollution.**
    Ans: Waterpollutionisthecontaminationofwaterbodies,usuallyasaresultofhuman activities. Water bodies include for example lakes, rivers, oceans, aquifers and groundwater.Waterpollutionresultswhencontaminantsareintroducedintothenatural environment.
24. **When will you call a water bodypolluted?**

Ans: Polluted water may be turbid, foul smelling, and may contain number of pathogens, heavy metals, oils etc.

Even a small quantity of about 0.1% impurities in water, can make it unfit for human consumption

25. **Why do you think the amount of DDT is maximum inbirds?**

- **Ans:** 

    i. Afewsubstancesusuallypresentinindustrialwastewaterscanundergo biologicalmagnification.
    ii. The phenomenon through which certain pollutants get accumulated in tissues in increasing concentration along the food chains (successive trophic levels) is called **Biological Magnification(Biomagnification.**
    iii. Manyofthepesticidesarenon-degradable.So,theygetaccumulatedin thetissuesoforganisms.Theyneithergetmetabolisednorareexcreted.
    iv. They are passed on to the next trophic level. This is commonly seen in case of DDT andMercury.

- Here is an example of DDT in abovediagram.
- **S**prayingamarshtocontrolmosquitoeswillcausetraceamountsofDDTtoaccumulate in the cells of microscopic aquatic organisms, the[plankton], in themarsh.

- In feeding on the plankton, filter-feeders, like clams and some fish, harvest DDT as well as food. (Concentrations of DDT 10 times greater than those in the plankton have been measured inclams.)
- The process of concentration goes right on up the food chain from one trophic level to the next.
- Gulls,whichfeedonclams,mayaccumulateDDTto40ormoretimestheconcentration in their prey. This represents a 400-fold increase in concentration along the length of this short foodchain.
- There is abundant evidence that some carnivores at the ends of longer food chains(e.g. ospreys, pelicans, falcons, and eagles) suffered serious declines in fecundity and hence in population size because of this phenomenon in the years before use of DDT was banned.

26. Explain BOD and its effects on aquaticecosystem.

Ans: It is possible to estimate biodegradable organic matter in sewage water by measuring **Biochemical Oxygen Demand (BOD)**. It is the amount of dissolved oxygen required by microorganisms for decomposing the organic matter present in water. It is expressed in milligram of oxygen per litre (mg/L) of water. High BOD indicates intense level ofmicrobial pollution.

Microorganisms involved in biodegradation of organic matter in water body consume lot of dissolved oxygen. This results in sharp decline in oxygen level of water which leads to mortality of fish and other aquatic creatures.

Presence of large amount of nutrients in water causes excessive growth of **planktonic** free-floatingalgaespecially,bluegreenalgae.Thisiscalled**algalbloom**whichgivescolourtothe water bodies. Algal bloom often releases toxins in water. So, fish die due to toxicity. Quality of water deteriorates and becomes toxic for human beings and otheranimals.

27. Ecological sanitation is the need of the day. Justify.

Ans: Ecological sanitation(Ecosan)

i. **Ecological sanitation (Ecosan)** is an approach to sanitation provision which safely reuses excreta in agriculture. It reduces the need for chemical fertilizers.
ii. Ecosan toilet is a closed system that does not need water. It is an alternative to leach pit toilets in place where water is scarce or where there is risk of ground watercontamination.
iii. It is based on the principle of recovery and recycling of nutrients from excreta to create a valuable resource foragriculture.
iv. When the pit of an ecosan toilet fills up, it is closed andsealed.
v. After about 8-9 months, the faeces are completely composted to organicmanure.
vi. There are working EcoSan toilets in many areas of Gujrat, Kerala, Tamilnadu and SriLanka

28. Explain the phenomenon of biologicalmagnification.

i. **Ans:** A few substances usually present in industrial waste waters can undergo biological magnification.

ii. The phenomenon through which certain pollutants get accumulated in tissues in increasing concentration along the food chains (successive trophic levels) is called **Biological Magnification(Biomagnification.**

iii. Manyofthepesticidesarenon-degradable.So,theygetaccumulatedin thetissuesoforganisms.Theyneithergetmetabolisednorareexcreted. Theyarepassedontothenexttrophiclevel.ThisiscommonlyseenincaseofDDTand Mercury

29. **Why is it necessary to separate wet and drygarbage?**

Ans: Waste segregation refers to the **separation of dry** and **wet garbage**, whichpaves the way for other concepts of **waste** management like composting, recycling and incineration.Itsendgoalistoreduce**waste**fromlandfillsandeventually,preventland, water and airpollution

30. **What are the six R's to combat solidwaste?**

Ans: The **6 Rs**include: reinvent/rethink, refuse, reduce, reuse/repair, recycle, replace/rebuy.

31. **What is greenhouseeffect?**

Ans: The **greenhouse effect** is a natural process that warms the Earth's surface. When the Sun's energy reaches the Earth's atmosphere, some of it is reflected back to space and therest is absorbed and re-radiated by **greenhousegases**.

32. **Enlist greenhouse gases.**

Ans: CO_2, CO, CH, CFC, NO_2etc are green house gases

33. **How do you corelategreen house effect and global warming?**

34. **Ans:** Carbon dioxide (CO_2) and other **greenhouse gases** act like a blanket, absorbing IR radiation and preventing it from escaping into outer space. The net **effect** is the gradualheatingofEarth'satmosphereandsurface,aprocessknownas**globalwarming.**

35. **How pollution by domestic garbage can be controlled?**

Ans: Domestic garbage can be controlled by followingways:-

1. **Segregate et waste (biodegradable0 waste from dry weight (nondegradable waste)**
2. Say no to plastic bag. ...
3. Buy foods with less packaging....
4. Buy cosmetic in bulk. ...
5. Avoid packaged drinks....
6. Donate old unused items....
7. Reuse what you can....
8. Follow the recycle policies....
9. Handle hazardous wastes properly.

36. **Give effect of CFC on ozoneshield.**

Ans:

1. The ozone shield has been disturbed due to increased rate of ozone degradation by **Chlorofluorocarbon** (CFC). CFCs move upwards and reachstratosphere.
2. UV rays act on them and release Cl atoms. Cl degrades ozone releasing molecular oxygen. Cl atoms act ascatalyst.

3. So they remain in the stratosphere and continue the effect of ozonedegradation.
4. This results in ozonedepletion.
5. Although it occurs widely in the stratosphere, the depletion is particularly marked over the Antarctic region.
6. This has resulted in formation of large area of thinned ozone layer, commonly called **ozonehole.**

37. Give an account of possible effects of globalwarming.

Ans: Effects of global warming

- **i.** Melting glaciers, early snowmelt, and severe droughts will cause more dramatic water shortages and increase the risk ofwildfires.
- **ii.** Rising sea levels will lead to coastalflooding
- **iii.** Forests, farms, and cities will face troublesome new pests, heat waves, heavy downpours, and increased flooding. All those factors will damage or destroy agriculture andfisheries.
- **iv.** Disruption of habitats such as coral reefs and Alpine meadows could drive many plant and animal species toextinction.
- **v.** Allergies, asthma, and infectious disease outbreaks will become more common due to increased growth of pollen-producing ragweed, higher levels ofair pollution, and the spread of conditions favorable to pathogens andmosquitoes.

38. Plastic ban in Maharashtra is an essential step. Givereason.

Ans: 1. The state government, in a notification on March 23, had clamped down on the use of **plastic** in a bid to fight pollution caused due to its extensive use.

3. Thenotification**banned**manufacturing,use,sale,distributionandstorageof **plastic**materials such as **bags**, spoons, plates and other disposable items.
4. **Plastic bags** tend to disrupt the **environment** in a seriousway.
5. They get into soil and slowly release **toxic**chemicals.
6. They eventually break down into the soil, with the unfortunate result being that animals eat them and often choke and die
7. **Plastic** is a petroleum-based material, and when burned it's like any other fossil fuel: it releases climate**pollution**.
8. This in turn leads to rising sea levels, increased ocean and **air** toxicity, and destruction of coral reefs and other marine life.

By 2050 **plastic** could emit 56 billion tonnes of greenhouse gas emissions, as **much** as 14 percent of the earth's remaining carbon budget. By 2100 it will emit 260 billion tonnes, more than half of the carbon budget

39. Find out more about artificial glaciers.

Ans: Articial glaciers

- **i.** In order to overcome the problem of global warming, ChewangNorphel, Ice Man of India has built 13 artificialglaciers.
- **ii.** He is an Indian civil engineer from Ladakh. Norphel noticed a small stream had frozen solid, under the shade of a group of poplar trees, though it flowed freely elsewhere in hisyard.

iii. The reason for this phenomenon : The flowing water was moving quickly to freeze while the sluggish trickle of water beneath the trees, was slow enough tofreeze.

iv. Based on this, he created artificial glaciers by diverting a river into a valley, slowing the stream by constructingchecks.

v. Theartificialglaciersincreasethegroundwaterrecharge,rejuvenating the spring and providing water forirrigation.

vi. He constructed them at lower elevations, so that they melt earlier expanding the growingseason.

40. Collect information about colour code used for biomedical waste.

Ans:

YELLOW BAGS	RED BAGS	BLUE BAGS	BLACK CARBOY
Infectious waste, bandages, gauze, cotton or any other objects in contact with body fluids, human body parts, placenta etc.	Plastic waste such as catheters,injection syringes, tubings, iv bottles	All types of glass bottles and broken glass articles, outdated & discarded medicines	Needles without syringes, blades, sharps and all metal articles.

41. What is Chipkomovement?

- **Ans:** ChipkoMovement:
- History of people's participation in India can be traced back to1731.
- The ministers of the king of Jodhpur in Rajasthan tried to cut forest to procure wood for a newpalace.
- LocalBishnoicommunityknownforpeacefulco-existencewithnature,opposedking's men.
- A Bishnoi woman Amrita Devi hugged the trees and lost her life in an attempt for protecting the forest. Her three daughters and hundreds of other Bishnois too lost their lives

42. How Jhumcultication has led to deforestation in recentyears?

Ans: i. **Slash and burn agriculture** commonly called **Jhum Cultivation** in north eastern parts of India, where farmers cut down trees of the forest and burn the plant remains.

ii. The ash is used as fertilizer and the land is used for farming and cattlegrazing.

iii. After cultivation, the area is left for several years so as to allow itsrecovery

43. Comment on deforestation status of the world and its majoreffects.

Ans; 1. Deforestation is conversion of forest area into non-forest area.

ii. According to an estimate, almost 40% forests are lost in the tropics and only 1% in temperateregion.

iii. The scenario of deforestation is grim inIndia.

iv. At the beginning of 20th century, forest cover was about 30% inIndia.

v. By the end of the century, it got reduced to 19.4%.

vi. TheNationalForestPolicy1988ofIndiahasrecommended33%forestcoverfortheplains and 67% for thehills.

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